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Inventor(s)

Rosenberg; Andrew et al.

Speculative ASR Decoding to Reduce Overall Latency of Speech Applications

Abstract

A method includes receiving a training utterance including a sequence of audio frames and a corresponding ground-truth transcription and truncating a suffix portion from the sequence of audio frames to provide a prefix sequence of the audio frames. The method also includes processing, using a speech recognition model, the prefix sequence of audio frames to provide a partial speech recognition hypothesis and processing, using a prefix embedding multi-head attention layer, a sequence of prefix encodings to generate a prefix prompt embedding. The method also includes processing, using a LM, the partial speech recognition hypothesis conditioned on the prefix prompt embedding to generate a speculated speech recognition (SSR) hypothesis and determining a training loss based on the SSR hypothesis and the ground-truth transcription. The method also includes fine-tuning parameters of the prefix embedding multi-head attention layer.

Inventors: Rosenberg; Andrew (Brooklyn, NY), Ramabhadran; Bhuvana (Mt. Kisco, NY), Yusuf; Bolaji (Mountain View, CA), Baskar; Murali Karthick (Mountain View, CA)

Applicant: Google LLC (Mountain View, CA)

Family ID: 94871233

Assignee: Google LLC (Mountain View, CA)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This U.S. patent application claims priority under 35 U.S.C. § 119 (e) to U.S. Provisional Application 63/559,668, filed on Feb. 29, 2024. The disclosure of this prior application is considered part of the disclosure of this application and is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] This disclosure relates to speculative ASR decoding to reduce overall latency of speech applications.

BACKGROUND

[0003] The experience of users interacting with an automatic speech recognition (ASR) system is colored by its latency, characterizing how quickly the ASR system is able to respond to user requests, in addition to speech recognition accuracy. An ASR system which can respond quickly to user queries is generally preferred to slower ASR systems with similar accuracy. While many training objectives have been developed to encourage early-output symbol emission to reduce latency as close to zero without incurring significant degradation in speech recognition accuracy, the best case scenario is for these ASR models to finish transcribing audio just as a user finishes speaking. However, in digital assistant interactions where speech recognition of a spoken query is only the first step in user interaction, speech recognition is typically followed by some form of natural language processing (NLP) to ascertain an action specified by the transcription of the query for the digital assistant to perform. Therefore, even if latency of the ASR model for generating the transcription were to reach zero, the overall end-to-end latency experienced by the user until the digital assistant performs the action, would still be above zero. Speculative speech recognition (SSR) empowers conventional speech recognition systems with speculation capabilities to allow speech recognition to run ahead of input audio characterizing an utterance spoken by a user before the user finishes speaking the utterance.

SUMMARY

[0004] One aspect of the disclosure provides a computer-implemented method executed on data processing hardware that causes the data processing hardware to perform operations that include receiving a training utterance including a sequence of audio frames characterizing an utterance and a corresponding ground-truth transcription of the utterance and truncating a suffix portion from the sequence of audio frames to provide a prefix sequence of the audio frames that characterizes a prefix portion of the utterance. The operations also include processing, using a speech recognition model including an encoder and a decoder, the prefix sequence of the audio frames to generate a partial speech recognition hypothesis for the prefix portion of the utterance, and processing, using a prefix embedding multi-head attention layer, a sequence of prefix encodings encoded by the encoder of the speech recognition model from the prefix sequence of audio frames to generate a prefix prompt embedding. The operations also include processing, using a language model (LM), the partial speech recognition hypothesis generated by the speech recognition model conditioned on the prefix prompt embedding to generate a speculated speech recognition hypothesis for the truncated suffix portion of the sequence of audio frames. The operations also include determining a training loss based on the speculated speech recognition hypothesis and the ground-truth

transcription of the utterance and fine-tuning, using the training loss, parameters of the prefix embedding multi-head attention layer.

[0005] Implementations of the disclosure may include one or more of the following optional features. In some implementations, the operations also include obtaining a sequence of trainable query vectors, wherein processing the sequence of prefix encodings encoded by the encoder to generate the prefix prompt embedding includes processing, using the prefix embedding multi-head attention layer, the sequence of prefix encodings and the sequence of trainable query vectors to generate the prefix prompt embedding. In these implementations, the trainable query vectors may include soft prompt query vectors and fine-tuning further includes fine-tuning the trainable query vectors.

[0006] In some examples, fine-tuning further includes performing parameter-efficient fine-tuning (PEFT) to only update a subset of existing or newly added parameters of the LLM based on the training loss. Here, the LM may be pre-trained and include a plurality of multi-head attention blocks, the pre-trained LM may be modified to add two low-rank projection matrices to each multi-head attention block, and performing PEFT may include fine-tuning only the parameters of the low-rank projection matrices added to the pre-trained LM while existing parameters of the pre-trained LM remain fixed.

[0007] In some implementations, the encoder of the speech recognition model includes a pre-trained audio encoder comprising a plurality of multi-head attention layers each including a multi-head attention mechanism and parameters of the pre-trained audio encoder are held fixed. In these implementations, the multi-head attention layers may include Conformer layers. Further, the LLM may include a plurality of multi-head attention layers that may include Transformer layers. Additionally, the speech recognition model may include a recurrent neural network-transducer architecture.

[0008] Another aspect of the disclosure provides a system including data processing hardware and memory hardware in communication with the data processing hardware and storing instructions that when executed on the data processing hardware causes the data processing hardware to perform operations that include receiving a training utterance including a sequence of audio frames characterizing an utterance and a corresponding ground-truth transcription of the utterance and truncating a suffix portion from the sequence of audio frames to provide a prefix sequence of the audio frames that characterizes a prefix portion of the utterance. The operations also include processing, using a speech recognition model including an encoder and a decoder, the prefix sequence of the audio frames to generate a partial speech recognition hypothesis for the prefix portion of the utterance, and processing, using a prefix embedding multi-head attention layer, a sequence of prefix encodings encoded by the encoder of the speech recognition model from the prefix sequence of audio frames to generate a prefix prompt embedding. The operations also include processing, using a language model (LM), the partial speech recognition hypothesis generated by the speech recognition model conditioned on the prefix prompt embedding to generate a speculated speech recognition hypothesis for the truncated suffix portion of the sequence of audio frames. The operations also include determining a training loss based on the speculated speech recognition hypothesis and the ground-truth transcription of the utterance and fine-tuning, using the training loss, parameters of the prefix embedding multi-head attention layer.

[0009] Implementations of the disclosure may include one or more of the following optional features. In some implementations, the operations also include obtaining a sequence of trainable query vectors, wherein processing the sequence of prefix encodings encoded by the encoder to generate the prefix prompt embedding includes processing, using the prefix embedding multi-head attention layer, the sequence of prefix encodings and the sequence of trainable query vectors to generate the prefix prompt embedding. In these implementations, the trainable query vectors may include soft prompt query vectors and fine-tuning further includes fine-tuning the trainable query vectors.

[0010] In some examples, fine-tuning further includes performing parameter-efficient fine-tuning (PEFT) to only update a subset of existing or newly added parameters of the LLM based on the training loss. Here, the LM may be pre-trained and include a plurality of multi-head attention blocks, the pre-trained LM may be modified to add two low-rank projection matrices to each multi-head attention block, and performing PEFT may include fine-tuning only the parameters of the low-rank projection matrices added to the pre-trained LM while existing parameters of the pre-trained LM remain fixed.

[0011] In some implementations, the encoder of the speech recognition model includes a pre-trained audio encoder comprising a plurality of multi-head attention layers each including a multi-head attention mechanism and parameters of the pre-trained audio encoder are held fixed. In these implementations, the multi-head attention layers may include Conformer layers. Further, the LLM may include a plurality of multi-head attention layers that may include Transformer layers. Additionally, the speech recognition model may include a recurrent neural network-transducer architecture.

[0012] The details of one or more implementations of the disclosure are set forth in the accompanying drawings and the description below. Other aspects, features, and advantages will be apparent from the description and drawings, and from the claims.

Description

DESCRIPTION OF DRAWINGS

[0013] FIG. 1 is a schematic view of a system executing a speculative speech recognition model.

[0014] FIG. 2 is a schematic view of an example speech recognition model having a recurrent neural network-transducer architecture.

[0015] FIG. 3 is a schematic view of an example training process for training the speculative speech recognition model of FIG. 1.

[0016] FIG. 4 is a schematic view of an example Alignment With Subsequence Edit Distance (AWSED) procedure executed by the training process of FIG. 3

[0017] FIG. 5 is a flowchart of an example arrangement of operations for a computer-implemented method of training the speculative speech recognition model.

[0018] FIG. 6 is a schematic view of an example computing device that may be used to implement the systems and methods described herein.

[0019] Like reference symbols in the various drawings indicate like elements.

DETAILED DESCRIPTION

[0020] The experience of users interacting with an automatic speech recognition (ASR) system is colored by its latency, characterizing how quickly the ASR system is able to respond to user requests, in addition to speech recognition accuracy. An ASR system which can respond quickly to user queries is generally preferred to slower ASR systems with similar accuracy. While many training objectives have been developed to encourage early-output symbol emission to reduce latency as close to zero without incurring significant degradation in speech recognition accuracy, the best case scenario is for these ASR models to finish transcribing audio just as a user finishes speaking. However, in digital assistant interactions where speech recognition of a spoken query is only the first step in user interaction, speech recognition is typically followed by some form of natural language processing (NLP) to ascertain an action specified by the transcription of the query for the digital assistant to perform. Therefore, even if latency of the ASR model for generating the transcription were to reach zero, the overall end-to-end latency experienced by the user until the digital assistant performs the action, would still be above zero.

[0021] Prefetching techniques provide a template for reducing the end-to-end latency further by hinging on the observation that there is a delay between an ASR system's emission of a last output

symbol recognized and being able to confidently determine that the user has finished speaking an utterance. Namely, ASR hypotheses output by the ASR system are sent downstream for NLP processing as soon as a token is emitted without waiting for the end-of-utterance confirmation such that downstream NLP computation can commence immediately. Thus, these prefetching techniques mitigate end pointing latency in exchange for extra computation overhead.

[0022] Large language models (LLMs) have become more popular in recent years. A number of recent studies have explored the use of LLMs for spoken language understanding and fine-tuning the LLMs on audio data to perform speech-to-text question answering tasks. These models were able to answer text questions in response to input audio in a direct manner. Speculative speech recognition (SSR) empowers conventional speech recognition systems with speculation capabilities to allow speech recognition to run ahead of input audio characterizing an utterance spoken by a user before the user finishes speaking the utterance.

[0023] Implementations herein are directed towards a SSR model and methods of training and executing the SSR model to accurately generate a full transcription of an utterance spoken by a user before the user has finished speaking the utterance. Specifically, implementations are directed toward the SSR model including an automated speech recognition (ASR) model and a large language model (LLM) modified such that the LLM receives, as input, a partial speech recognition hypothesis output by the ASR model for a prefix portion of an utterance and an audio-dependent prefix prompt embedding generated by a prefix embedding multi-head attention layer, and generates, as output, a speculated speech recognition hypothesis for a suffix portion of the utterance. Here, the prefix portion of the utterance is spoken by a user before the user speaks the suffix portion of the utterance and the audio-dependent prefix prompt embedding generated by the prefix embedding multi-head attention layer is based on a sequence of prefix encodings encoded by an audio encoder of the ASR model from a sequence of audio frames characterizing the prefix portion of the utterance. Thereafter, a downstream component may proactively process a transcription of the utterance formed by the partial speech recognition hypothesis output by the ASR model for the prefix portion of the utterance and the SSR hypothesis output by the LLM for the suffix portion of the utterance to identify an action for an application (e.g., a digital assistant) to perform before the user completes speaking the suffix portion of the utterance. Thus, the ability of the SSR model to accurately generate full transcriptions of utterances before the complete utterance is finished can permit initiation of downstream query interpretation (e.g., NLP) operations early to reduce the end-to-end latency of fulfilling tasks specified by user utterances.

[0024] FIG. 1 illustrates an example system **100** whereby a user **10** may interact with a computing device, such as a user device **110**, through voice input. The user device **110** (also referred to generally as a device **110**) is configured to capture sounds (e.g., streaming audio data) from one or more users **10**. Here, the streaming audio data may refer to an utterance **106** spoken by the user **10** that functions as an audible prompt/query, a command for the user device **110**, or an audible communication captured by the user device **110**. Speech-enabled systems of the user device **110** may field the query or command by answering the query and/or causing the command to be performed/fulfilled by one or more downstream applications. For instance, in the example shown, the user **10** interacts with a digital assistant **50** of the user device **110** that uses a speculative speech recognition (SSR) model **120**. The digital assistant **50** displays a digital assistant interface **118** on a screen of the user device **110** to depict a conversation between the user **10** and the digital assistant **50**.

[0025] The user device **110** may correspond to any computing device associated with the user **10** and capable of receiving audio data. Some examples of user devices **110** include, but are not limited to, mobile devices (e.g., mobile phones, tablets, laptops, etc.), computers, wearable devices (e.g., smart watches), smart appliances, internet of things (IoT) devices, vehicle infotainment systems, smart displays, smart speakers, etc. The user device **110** includes data processing hardware **112** and memory hardware **114** in communication with the data processing hardware **112**

and stores instructions, that when executed by the data processing hardware **112**, causes the data processing hardware **112** to perform one or more operations. The user device **110** further includes an audio system **116** with an audio capture device (e.g., microphone) **116**, **116a** for capturing and converting the utterances **106** spoken by the user **10** into electrical signals and a speech output device (e.g., speaker) **116**, **116b** for communicating an audible audio signal (e.g., as output audio data from the user device **110**). That is, the audio capture device **116a** may convert the utterances **106** spoken by the user **10** into a sequence of audio frames **102**. While the user device **110** implements a single audio capture device **116a** in the example shown, the user device **110** may implement an array of audio capture devices **116a** without departing from the scope of the present disclosure, whereby one or more capture devices **116a** in the array may not physically reside on the user device **110**, but be in communication with the audio system **116**.

[0026] The user device **110** communicates with a remote system **140** via a network **130**. The remote system **140** may be a distributed system (e.g., cloud computing environment) having scalable elastic resources. The resources include computing resources (e.g., data processing hardware) **142** and/or storage resources (e.g., memory hardware) **144**. Additionally or alternatively, the remote system **104** may be a centralized system. The network **130** may be wired, wireless, or a combination thereof, and may include private networks and/or public networks, such as the Internet.

[0027] The SSR model **120** may execute on the user device **110**, the remote system **140**, or some combination thereof. The SSR model **120** is configured to receive, as input, a sequence of audio frames **102** characterizing a prefix portion **106a** of an utterance **106** spoken by a user before the user **10** speaks a suffix portion of the utterance **106**, and generate, as output, both a partial speech recognition hypothesis for the prefix portion **106a** of the utterance **106** and a speculated speech recognition hypothesis **180b** for the suffix portion of the utterance **106** before the user completes speaking the suffix portion of the utterance **106**. A transcription **180** of the utterance **106** is formed by the partial speech recognition hypothesis **180a** and the speculated speech recognition hypothesis **180b**. Thereafter, a downstream query interpreter **190** processes the transcription **180** of the utterance **106** to identify an action for the digital assistant **50** (or other software application) to perform. The query interpreter **190** may include a downstream natural language processing (NLP) and/or natural language understanding (NLU) module configured to identify the action proactively before the user completes speaking the utterance. Optionally, the query interpreter **190** may include a language model (LM) **160** of the SSR model **120**. After identifying the action, the query interpreter **190** may fulfill performance of the action and provide a response **192** for output from the user device **110** indicating fulfillment of the action. As will become apparent, the SSR model **120** is trained to generate the speculative hypothesis **180b** for the suffix portion of the utterance **106** before the user **10** speaks the suffix portion of the utterance **106** based on the partial speech recognition hypothesis **180a** generated for the prefix portion **106a** of the utterance **106** and an audio-dependent prefix prompt embedding **175** generated by a prefix embedding multi-head attention layer **170** based on the sequence of audio frames **102** characterizing the prefix portion **106a** of the utterance **106**.

[0028] In some examples, the utterances **106** spoken by the user **10** correspond to spoken prompts. As such, utterances **106** may be interchangeably referred to as “spoken prompts” herein. Spoken prompts **106** may include any query, command, or other audible communication captured by the user device **110** (e.g., any command or query spoken by the user **10**).

[0029] In the example shown, the user speaks “Who is the pre-” as the prefix portion **106a** of the utterance **106** and the SSR model **120** generates the SSR hypothesis **180b** of “-sident of Liberia” as the suffix portion of the utterance **106** before the user speaks the suffix portion of the utterance. Here, the transcription **180** formed by the partial speech recognition hypothesis **180a** and the SSR hypothesis **180b** may be a question spoken by the user **10** whereby the digital assistant **50** obtains “Joseph Boakai is the present of Liberia” as a response **192** that answers the question. After the

query interpreter **190** identifies the action, the present disclosure is not limited to how the digital assistant **50** or other downstream application fulfills the action to obtain the response **192**. For instance, the digital assistant **50** may input the transcription **180** as a search query to a search engine to obtain the response **192** from the search engine. In other examples, the LM **160** may process the transcription **180** as a textual prompt to generate the response **192**. The query interpreter **190** may identify the action as a command for an automation device to perform an operation such as “turning lights on” or “playing a music playlist”, whereby the digital assistant **50** may provide a response **192** indicating the operation has been performed by the automation device.

[0030] The SSR model **120** includes a speech recognition model **200**, a prefix embedding multi-head attention layer **170**, and the LM **160**. The speech recognition model **200** includes an audio encoder **210** and a speech decoder **250**. In some implementations, the speech recognition model **200** includes a recurrent neural network-transducer (RNN-T) model architecture or other frame alignment-based transducer model which adheres to latency constraints associated with interactive applications. Referring to FIG. 2, an example frame alignment-based transducer model **200** includes a Recurrent Neural Network-Transducer (RNN-T) model architecture is shown. The use of the RNN-T model architecture is exemplary, and the frame alignment-based transducer model **200** may include other architectures such as transformer-transducer and conformer-transducer model architectures among others. The RNN-T model **200** provides a small computational footprint and utilizes less memory requirements than conventional ASR architectures, making the RNN-T model architecture suitable for performing speech recognition entirely on the user device **110** (e.g., no communication with a remote server is required). The RNN-T model **200** includes the audio encoder **210**, a prediction network **220**, and a joint network **230**. The audio encoder **210**, which is roughly analogous to an acoustic model (AM) in a traditional ASR system, includes a stack of self-attention layers (e.g., Conformer or Transformer layers) or a recurrent network of stacked Long Short-Term Memory (LSTM) layers. For instance, the audio encoder **210** reads a sequence of d-dimensional feature vectors (e.g., audio frames **102** (FIG. 1)) $x = (x_{\text{sub}.1}, x_{\text{sub}.2}, \dots, x_{\text{sub}.T})$, where $x_{\text{sub}.t} \in \mathbb{R}^d$, and produces at each output step a higher-order feature representation. This higher-order feature representation is denoted as $h_{\text{sub}.1.\text{sup}.enc}, \dots, h_{\text{sub}.T.\text{sup}.enc}$ and may be interchangeably referred to as an audio encoding **212**.

[0031] Similarly, the prediction network **220** is also an LSTM network, which, like a language model (LM), processes the sequence of non-blank symbols output by a final Softmax layer **240** so far, $y_{\text{sub}.0}, \dots, y_{\text{sub}.ui-1}$, into a dense representation $p_{\text{sub}.u.\text{sub}.i}$. Finally, with the RNN-T model architecture, the representations produced by the encoder and prediction/decoder networks **210**, **220** are combined by the joint network **230**. The prediction network **220** may be replaced by an embedding look-up table to improve latency by outputting looked-up sparse embeddings in lieu of processing dense representations.

[0032] The joint network then predicts $P(y_{\text{sub}.i} | x_{\text{sub}.t.\text{sub}.i}, y_{\text{sub}.0}, \dots, y_{\text{sub}.u.\text{sub}.i-1})$, which is a distribution over the next output symbol. Stated differently, the joint network **230** generates, at each output step (e.g., time step), a probability distribution over possible speech recognition hypotheses. Here, the “possible speech recognition hypotheses” correspond to a set of output labels each representing a symbol/character in a specified natural language. For example, when the natural language is English, the set of output labels may include twenty-seven (27) symbols, e.g., one label for each of the 26-letters in the English alphabet and one label designating a space. Accordingly, the joint network **230** may output a set of values indicative of the likelihood of occurrence of each of a predetermined set of output labels. This set of values can be a vector and can indicate a probability distribution over the set of output labels. In some cases, the output labels are graphemes (e.g., individual characters, and potentially punctuation and other symbols), but the set of output labels is not so limited. For example, the set of output labels can include wordpieces, phonemes, and/or entire words, in addition to or instead of graphemes. The output distribution of the joint network **230** can include a posterior probability value for each of the different output

labels. Thus, if there are 100 different output labels representing different graphemes or other symbols, the output $y_{sub.i}$ of the joint network **230** can include 100 different probability values, one for each output label. The probability distribution can then be used to select and assign scores to candidate orthographic elements (e.g., graphemes, wordpieces, and/or words) in a beam search process (e.g., by the Softmax layer **240**) for determining the transcription **120**.

[0033] The Softmax layer **240** may employ any technique to select the output label/symbol with the highest probability in the distribution as the next output symbol predicted by the RNN-T model **200** at the corresponding output step. In this manner, the RNN-T model **200** does not make a conditional independence assumption, rather the prediction of each symbol is conditioned not only on the acoustics but also on the sequence of labels output so far. The RNN-T model **200** does assume an output symbol is independent of future acoustic frames **102**, which allows the RNN-T model to be employed in a streaming fashion.

[0034] In some examples, the audio encoder **210** of the RNN-T model **200** includes a stack of self-attention layers/blocks, such as conformer layers/blocks. In some examples, the number of conformer layers/blocks in the audio encoder is equal to 17 with 512-dimensional layers. The audio encoder **210** may include 100 million parameters. Here, each conformer block includes a series of multi-headed self attention, depth wise convolution and feed-forward layers. The stack of self-attention layers/blocks may include transformer layers/blocks in other examples. The prediction network **220** may have one 512-dimensional LSTM layer. Alternatively, the prediction network **220** may include a stack of transformer or conformer blocks, or an embedding look-up table in lieu of LSTM layers. Finally, the joint network **230** may include two (2) feedforward layers with a 512-dimensional intermediate layer. The Softmax layer **240** may be composed of a unified word piece or grapheme set that is generated using all unique word pieces or graphemes in a plurality of training data sets. The Softmax layer **240** may include a 1024-dimensional layer corresponding to 1,024 wordpiece targets. The prediction network **220**, the joint network **230**, and the Softmax layer **240** may collectively form an RNN-T decoder **250** of the RNN-T model **200**. Thus, the speech decoder **250** of FIG. 1 may include the RNN-T decoder **250**.

[0035] Referring back to FIG. 1, the audio encoder **210** is configured to receive, as input, the sequence of audio frames **102** characterizing the prefix portion **106a** of the utterance **106** and generate, as output, a corresponding sequence of prefix encodings **212**. The speech decoder **250** (e.g., RNN-T decoder) is configured to process the corresponding sequence of the prefix encodings **212** to generate the partial speech recognition hypothesis **180a** for the prefix portion **106a** of the utterance **106**.

[0036] The sequence of audio frames **102** may include an input sequence of mel-frequency spectrogram frames. In some examples, the audio encoder **210** operates in a streaming manner. That is, for each respective audio frame **102** in the sequence of audio frames **102**, the audio encoder **210** generates a corresponding prefix encoding **212** and passes the corresponding prefix encoding **212** to the speech decoder **250** of the speech recognition model **200**. As such, at each time step (e.g., output step) of a plurality of time steps, the audio encoder **210** generates a corresponding prefix encoding **212**. The audio encoder **210** may additionally or alternatively operate in a non-streaming mode and process look-ahead or right context audio frames **102** when generating a prefix encoding **212** for a corresponding audio frame **102**. In some implementations, the audio encoder **210** includes a cascaded audio encoder that includes a causal encoder and a non-causal encoder stacked on top of the causal encoder.

[0037] The prefix embedding multi-head attention layer **170** of the SSR model **120** is configured to receive, as input, the sequence of prefix encodings **212** encoded by the audio encoder **210** and a query prompt Q characterized by a sequence of M trainable query vectors **172**, $q_{sub.1}, \dots, q_{sub.M}$, and generate, as output, a prefix prompt embedding ($A_{sup.P}$) **175**. The M trainable query vectors **172** may include soft prompt query vectors. Here, the keys and values of the prefix embedding multi-head attention layer **170** are the projections of the prefix encodings **212** output

from the audio encoder **210**, while the queries of the prefix embedding multi-head attention layer **170** are the projections of the of the M trainable query vectors ($q.sub.1, \dots, q.sub.M$) **172**. Thus, the prefix embedding multi-head attention layer **170** processes the sequence of prefix encodings **212** conditioned on the sequence of the M trainable query vectors ($q.sub.1, \dots, q.sub.M$) **172** to generate the prefix prompt embedding ($A.sup.P$) **175** as a fixed-length summary of the prefix encodings **212** represented by M audio-dependent embeddings $a.sub.1, \dots, a.sub.M$. Notably, the prefix encodings **212** are rich in lexical content since they correspond to encodings directly used by the speech recognition model **200** for speech recognition. Moreover, the prefix prompt embedding ($A.sup.P$) **175** is provided as a prefix to the LM **160** to reduce, and keep fixed, a computational cost of the subsequent decoding performed by the LM **160** compared to processing the prefix encodings **212** directly. In some examples, M is equal to 64 trainable query vectors **172** (equivalent in length to 2.56 seconds of audio frames **102**) of 1,024 dimensions and the prefix embedding multi-head attention layer **170** includes a single 1,024-dimensional cross-attention layer with four heads for computing the prefix prompt embedding **175**.

[0038] The LM **160** processes the partial speech recognition hypothesis **180a** generated by the speech recognition model **200** conditioned on the prefix prompt embedding ($A.sup.P$) **175** to generate the SSR hypothesis **180b** for the suffix portion of the utterance **106**. Moreover, the LM **160** is finetuned with low-rank (LoRA) adapters **161** along with the trainable query vectors **172** and parameters of the prefix embedding multi-head attention layer **170** to teach the LM **160** to learn how to speculate ideal suffix tokens to complete provide the best SSR hypothesis **180b** for the suffix portion of the utterance **106** that completes the partial speech recognition hypothesis **180a** for the prefix portion of the utterance **106**. While examples herein disclose using LoRA adapters **161** as a parameter-efficient finetuning (PEFT) technique for the LM **160**, the LM **160** may be finetuned using other PEFT techniques without departing from the scope of the present disclosure, such as adapters, residual adapters, or other sub-models having trainable parameters for adapting the pertained LM **160** on the SSR task. The SSR model **120** uses the speech recognition model **200** to predict the partial speech recognition hypothesis ($\hat{Y}.sup.P=y.sub.1, \dots, y.sub.r$) **180a** from the sequence of audio frames ($X.sup.P$) **102** characterizing the prefix portion **106a** of the utterance and the LM **160** is used to predict the SSR hypothesis $\hat{Y}.sup.S=y.sub.r+1, \dots, y.sub.U$ **180b** by conditioning the LM **160** on the audio-dependent prefix prompt embedding ($A.sup.P$) **175** and the partial speech recognition hypothesis **180a** as follows:

$$[00001] P(Y | X^P) = P(Y^P | X^P) \cdot \prod_{u=1}^U P_{\phi_e, \zeta, Q}(y_u | y_{<u}, X^P) \quad (1)$$

Where ζ denotes the parameters of the prefix embedding multi-head attention layer **170**, θ denotes the parameters of the LM **160**, $\phi.sub.e$ denotes the parameters of the audio encoder **210** of the speech recognition model **200**, and Q denotes the trainable query vectors **172**.

[0039] Notably, the LM **160** generates the SSR hypothesis **180b** for the suffix portion of the utterance **106** before the user speaks, and thus before the SSR model **120** processes any audio characterizing, the suffix portion of the utterance **106**. In the example, the SSR hypothesis **180b** for the suffix portion of the utterance **106** is “-sident of Liberia”. Accordingly, the transcription **180** is “who is the president of Liberia” formed by the partial speech recognition hypothesis **180a** (“who is the pre-”) and the SSR hypothesis **180b** (“-sident of Liberia”). In some examples, the LM **160** generates an n-best list of candidate SSR hypotheses and a Softmax of the LM **160** selects the SSR hypothesis **180b** from the n-best list of candidate SSR hypotheses. For instance, in the example shown in FIG. **1**, the n-best list of candidate SSR hypotheses for the suffix portion of the utterance **106b** may include “-sident of Liberia”, “-sident's husband”, “-sent head of NASA”, “-ss secretary”.

[0040] The LM **160** may include a 100 million parameter LM decoder having eight (8) 1,024-dimensional multi-head attention layers, each split into 16 attention heads, and including a tied embedding-softmax with the same 1,024 word piece units as the output softmax layer **240** of the RNN-T decoder **250**. In some examples, the multi-head attention layers of the LM **160** include

Transformer layers.

[0041] Notably, while simply using a hybrid model that stacks the speech recognition model **120** in cascade with the LM **160** where the speech recognition model generates and feeds the partial speech recognition hypothesis **180a** for the prefix portion **106a** of the utterance **106** to the LM will provide speculations of likely completions for the hypotheses of the suffix portion of the utterance **106**, such an architecture is limited in that the LM **160** is pretrained for generic text completion, and, consequently, does not account for the idiosyncrasies of operating on a speech recognition output. Specifically, the LM **160** would not account for the fact that the only input is the partial speech recognition hypothesis **180a** generated by the speech recognition model **200** and may contain errors. Stated differently, the LM **160** would not consider valuable information contained in the speech frames **102** which may be lost in the text-only partial speech recognition hypothesis **180a**, but would be useful for speculation. Accordingly, the SSR model **120** disclosed herein prepends the prefix prompt embedding **175** as an audio-dependent soft prompt to the partial speech recognition hypothesis **180a** as input to the LM **160**. As shown in FIG. 1, grey-shaded components of the SSR model **120** that include the trainable query vectors **172**, the prefix embedding multi-head attention layer **170**, and the LoRA adapters **161** are fine-tuned on the SSR task while parameters of the audio encoder **210**, speech decoder **250**, and LM **160** are held fixed.

[0042] Referring to FIG. 3, a training process **300** for fine-tuning the parameters $\Delta\theta$ of the LoRA adapters **161** of the LM **160** along with the tied embedding-softmax layer of the LM **160**, the parameters (ζ) of the prefix embedding multi-head attention layer **170**, and the soft prompt trainable query vectors (Q) **172** on a plurality of training utterances **106**, **106Ta-106T** to maximize a log-likelihood of predicting a ground-truth suffix portion (Y.sup.S) **106b** of each training utterance **106T**. Each training utterance **106T** includes a corresponding sequence of audio frames **102** paired with a corresponding ground-truth transcription **320**. For each training utterance **106T**, an audio truncator **310** truncates a suffix portion (X.sup.S) **102S** from the corresponding sequence of audio frames **102** to provide a prefix sequence of the audio frames (X.sup.P) **102P** that characterizes a prefix portion **106a** of the training utterance **106T**. For instance, the truncator **310** may truncate a fixed-length number of the last audio frames **102** in the sequence of audio frames **102**. In some examples, the fixed-length number is equal to a duration of about one (1) second of audio. Thereafter, the remaining prefix sequence of the audio frames **102P**.

[0043] The audio encoder **210** processes each audio frame in the prefix sequence of the audio frames (X.sup.P) **102P** to generate a corresponding sequence of prefix audio encodings **212** that is fed to both the speech decoder **250** and the prefix embedding multi-head attention layer **170**. The speech decoder **250** processes the sequence of prefix audio encodings **212** associated with the prefix portion of the training utterance to generate, as output, a partial speech recognition hypothesis ($\hat{Y}^{sup.P}$) **180a** for the prefix portion of the training utterance. The partial speech recognition hypothesis ($\hat{Y}^{sup.P}$) **180a** may be represented as a sequence of tokens (y.sub.1, . . . , y.sub.r) that transcribe the prefix portion of the training utterance. Simultaneously, the prefix embedding multi-head attention layer **170** receives the sequence of M trainable query vectors (q.sub.1, . . . , q.sub.M) **172** and processes the sequence of prefix encodings **212** conditioned on the sequence of M training query vectors **172** to generate a prefix prompt embedding (A.sup.P=a.sub.1 . . . , a.sub.M) **175**. Thereafter, the LM **160** modified by the LoRA adapters **161** processes the partial speech recognition hypothesis **180a** generated by the speech recognition model **200** conditioned on the prefix prompt embedding **175** to generate a SSR hypothesis (Y.sup.S) **180b** for the truncated suffix portion (X.sup.S) of the corresponding sequence of audio frames **102**. The LoRA adapters **161** may include a rank set depending on the type of training utterances the LoRA adapters **161** are finetuned on during the training process. For instance, the rank of the LoRA adapters **161** may be set equal to 10 for single-domain training utterances to provide 12 million trainable parameters to be finetuned for speculation. In other instances, the rank of the LoRA adapters **161** may be set equal to 50 for multi-domain training utterances to provide 19 million

trainable parameters to be finetuned for speculation.

[0044] The SSR hypothesis (Y.sup.S) **180b** may be represented as a sequence of tokens (y.sub.r+1, . . . , y.sub.U). In some examples, the LM **160** generates a plurality of candidate SSR hypothesis, such as an n-best list of candidate SSR hypotheses. A loss module **320** determines a training loss **325** based on the SSR hypothesis (Y.sup.S) **180b** (or the plurality of candidate SSR hypotheses) and the ground-truth transcription **320** of the training utterance **106T**. The training process **300** finetunes the LoRA adapters ($\Delta\theta$) **161** of the LM **160** along with the tied embedding-softmax layer of the LM **160**, the parameters (ζ) of the prefix embedding multi-head attention layer **170**, and the soft prompt trainable query vectors (Q) **172** based on the training loss **325**.

[0045] As the partial speech recognition hypothesis ($\hat{Y}$.sup.P) **180a** generated by the speech recognition model **200** for the prefix sequence of audio frames X.sup.P is represented by the sequence of tokens (y.sub.1, . . . , y.sub.r), determining the r for training, however, is not a trivial task. Consider, for example, a training utterance whose ground-truth transcription (Y) **320** is “i'd like to call my father”, but for which the speech recognition model **200** generates “i'd line call ma-” as the partial speech recognition hypothesis (PP) **180a** for the prefix sequence of audio frames X.sup.P. Due to the error generated by the speech recognition model **200**, a correct suffix portion of the ground-truth transcription is not clearly defined. Accordingly, it is necessary for the SSR model **120** to first be trained to determine what part of the ground-truth transcription **120** has been output by the speech recognition model **200**, perhaps erroneously, and what part remains to be speculated. To address this paradigm, FIG. 4 depicts the training process **300** executing an Alignment With Subsequence Edit Distance (AWSED) procedure **400**, whereby a Levenshtein distance is computed between the partial speech recognition hypothesis $\hat{Y}$.sup.P and all left-substrings of the ground-truth transcription Y. More specifically, by taking the left- and right-substring of the ground-truth transcription Y at some index v respectively as $Y_{\text{sub}:\text{v}} := (y_{\text{sub}.1}, \dots, y_{\text{sub}.v})$ and $Y_{\text{sub.v}:\text{v}} := (y_{\text{sub.v}+1}, \dots, y_{\text{sub}.U})$, then a target suffix, $Y_{\text{sup}.S} = Y_{\text{sub.v}^*}$, where:

$$[00002] \quad v^* = \underset{v}{\operatorname{argmin}} L(\hat{Y}^{\text{sup.P}}, Y_{\text{sub.v}}) \quad (2)$$

where $L(s_{\text{sub}.1}, s_{\text{sub}.2})$ is the Levenshtein distance between strings $s_{\text{sub}.1}$ and $s_{\text{sub}.2}$. Notably, rather computing $L(\hat{Y}^{\text{sup.P}}, Y_{\text{sub.v}})$ separately for each v, the training process **300** only requires one run of the AWSED procedure for computing the Levenshtein distance to get v^* because the last row of the matrix depicted in FIG. 4 of accumulated costs already contains $L(\hat{Y}^{\text{sup.P}}, Y_{\text{sub.v}}) \forall v$ and only the arg min of this row is required. In the event multiple indices are tied for the arg min, the AWSED procedure **400** will pick the leftmost one. Continuing with the example, the AWSED procedure **400** yields the resulting $Y_{\text{sup}.S}$ as “-my father”. The training process **300** may then use stochastic gradient descent to finetune. For each mini-batch B, the training process minimizes the cross-entropy by, with respect to the parameters ζ of the prefix embedding multi-head attention layer **170**, the soft prompt trainable query vectors Q, and the parameters $\Delta\theta$ of the LoRA adapters **161**:

$$[00003] \quad J_B = - \sum_{(X,Y) \in B} \sum_{u=v^*+1}^U \log P_{\theta, \zeta, Q} (y_u | X^P) \quad (3)$$

[0046] FIG. 5 includes a flowchart of an example arrangement of operations for a computer-implemented **500** of training a speculative speech recognition (SSR) model **120** to accurately generate a full transcription **180** of an utterance **106** spoken by a user **10** before the user **10** has finished speaking the utterance **106**. The operations for the method **500** can be described with reference to the training process **300** of FIG. 3 and the SSR model **120** of FIG. 1. The method **500** may execute on data processing hardware **610** (FIG. 6) using instructions stored on memory hardware **620** (FIG. 6) that may reside on the user device **110** and/or the remote system **140** of FIG. 1 each corresponding to a computing device **600** (FIG. 6).

[0047] At operation **502**, the method **500** includes receiving a training utterance **106**, **106T** including a sequence of audio frames **102** characterizing an utterance and a corresponding ground-

truth transcription **180** of the utterance. At operation **504**, the method **500** includes truncating (e.g., via truncator **310** (FIG. 3)) a suffix portion from the sequence of audio frames **102** to provide a prefix sequence of the audio frames **102P** that characterizes a prefix portion of the utterance. [0048] At operation **506**, the method includes processing, using a speech recognition model **200** including an encoder **210** and a decoder **250**, the prefix sequence of the audio frames **102P** to generate a partial speech recognition hypothesis **180a** for the prefix portion of the utterance **106**. At operation **508**, the method **500** includes processing, using a prefix embedding multi-head attention layer **170**, a sequence of prefix encodings **212** encoded by the encoder **210** of the speech recognition model **200** from the prefix sequence of audio frames **102P** to generate a prefix prompt embedding **175**.

[0049] At operation **510**, the method includes processing, using a language model (LM) **160**, the partial speech recognition hypothesis **180a** generated by the speech recognition model **200** conditioned on the prefix prompt embedding **175** to generate a speculated speech recognition hypothesis **180b** for the truncated suffix portion of the sequence of audio frames **102**.

[0050] At operation **512**, the method **500** includes determining a training loss **325** based on the speculated speech recognition hypothesis **180b** and the ground-truth transcription **320** of the utterance **106** and fine-tuning, using the training loss **325**, parameters ζ of the prefix embedding multi-head attention layer **170**.

[0051] FIG. **6** is a schematic view of an example computing device **600** that may be used to implement the systems and methods described in this document. The computing device **600** is intended to represent various forms of digital computers, such as laptops, desktops, workstations, personal digital assistants, servers, blade servers, mainframes, and other appropriate computers. The components shown here, their connections and relationships, and their functions, are meant to be exemplary only, and are not meant to limit implementations of the inventions described and/or claimed in this document.

[0052] The computing device **600** includes a processor **610**, memory **620**, a storage device **630**, a high-speed interface/controller **640** connecting to the memory **620** and high-speed expansion ports **650**, and a low speed interface/controller **660** connecting to a low speed bus **670** and a storage device **630**. Each of the components **610**, **620**, **630**, **640**, **650**, and **660**, are interconnected using various busses, and may be mounted on a common motherboard or in other manners as appropriate. The processor **610** can process instructions for execution within the computing device **600**, including instructions stored in the memory **620** or on the storage device **630** to display graphical information for a graphical user interface (GUI) on an external input/output device, such as display **680** coupled to high speed interface **640**. In other implementations, multiple processors and/or multiple buses may be used, as appropriate, along with multiple memories and types of memory. Also, multiple computing devices **600** may be connected, with each device providing portions of the necessary operations (e.g., as a server bank, a group of blade servers, or a multi-processor system).

[0053] The memory **620** stores information non-transitorily within the computing device **600**. The memory **620** may be a computer-readable medium, a volatile memory unit(s), or non-volatile memory unit(s). The non-transitory memory **620** may be physical devices used to store programs (e.g., sequences of instructions) or data (e.g., program state information) on a temporary or permanent basis for use by the computing device **600**. Examples of non-volatile memory include, but are not limited to, flash memory and read-only memory (ROM)/programmable read-only memory (PROM)/erasable programmable read-only memory (EPROM)/electronically erasable programmable read-only memory (EEPROM) (e.g., typically used for firmware, such as boot programs). Examples of volatile memory include, but are not limited to, random access memory (RAM), dynamic random access memory (DRAM), static random access memory (SRAM), phase change memory (PCM) as well as disks or tapes.

[0054] The storage device **630** is capable of providing mass storage for the computing device **600**.

In some implementations, the storage device **630** is a computer-readable medium. In various different implementations, the storage device **630** may be a floppy disk device, a hard disk device, an optical disk device, or a tape device, a flash memory or other similar solid state memory device, or an array of devices, including devices in a storage area network or other configurations. In additional implementations, a computer program product is tangibly embodied in an information carrier. The computer program product contains instructions that, when executed, perform one or more methods, such as those described above. The information carrier is a computer- or machine-readable medium, such as the memory **620**, the storage device **630**, or memory on processor **610**. [0055] The high speed controller **640** manages bandwidth-intensive operations for the computing device **600**, while the low speed controller **660** manages lower bandwidth-intensive operations. Such allocation of duties is exemplary only. In some implementations, the high-speed controller **640** is coupled to the memory **620**, the display **680** (e.g., through a graphics processor or accelerator), and to the high-speed expansion ports **650**, which may accept various expansion cards (not shown). In some implementations, the low-speed controller **660** is coupled to the storage device **630** and a low-speed expansion port **690**. The low-speed expansion port **690**, which may include various communication ports (e.g., USB, Bluetooth, Ethernet, wireless Ethernet), may be coupled to one or more input/output devices, such as a keyboard, a pointing device, a scanner, or a networking device such as a switch or router, e.g., through a network adapter.

[0056] The computing device **600** may be implemented in a number of different forms, as shown in the figure. For example, it may be implemented as a standard server **600a** or multiple times in a group of such servers **600a**, as a laptop computer **600b**, or as part of a rack server system **600c**.

[0057] Various implementations of the systems and techniques described herein can be realized in digital electronic and/or optical circuitry, integrated circuitry, specially designed ASICs (application specific integrated circuits), computer hardware, firmware, software, and/or combinations thereof. These various implementations can include implementation in one or more computer programs that are executable and/or interpretable on a programmable system including at least one programmable processor, which may be special or general purpose, coupled to receive data and instructions from, and to transmit data and instructions to, a storage system, at least one input device, and at least one output device.

[0058] These computer programs (also known as programs, software, software applications or code) include machine instructions for a programmable processor, and can be implemented in a high-level procedural and/or object-oriented programming language, and/or in assembly/machine language. As used herein, the terms “machine-readable medium” and “computer-readable medium” refer to any computer program product, non-transitory computer readable medium, apparatus and/or device (e.g., magnetic discs, optical disks, memory, Programmable Logic Devices (PLDs)) used to provide machine instructions and/or data to a programmable processor, including a machine-readable medium that receives machine instructions as a machine-readable signal. The term “machine-readable signal” refers to any signal used to provide machine instructions and/or data to a programmable processor.

[0059] The processes and logic flows described in this specification can be performed by one or more programmable processors, also referred to as data processing hardware, executing one or more computer programs to perform functions by operating on input data and generating output. The processes and logic flows can also be performed by special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application specific integrated circuit). Processors suitable for the execution of a computer program include, by way of example, both general and special purpose microprocessors, and any one or more processors of any kind of digital computer. Generally, a processor will receive instructions and data from a read only memory or a random access memory or both. The essential elements of a computer are a processor for performing instructions and one or more memory devices for storing instructions and data. Generally, a computer will also include, or be operatively coupled to receive data from or transfer

data to, or both, one or more mass storage devices for storing data, e.g., magnetic, magneto optical disks, or optical disks. However, a computer need not have such devices. Computer readable media suitable for storing computer program instructions and data include all forms of non-volatile memory, media and memory devices, including by way of example semiconductor memory devices, e.g., EPROM, EEPROM, and flash memory devices; magnetic disks, e.g., internal hard disks or removable disks; magneto optical disks; and CD ROM and DVD-ROM disks. The processor and the memory can be supplemented by, or incorporated in, special purpose logic circuitry.

[0060] To provide for interaction with a user, one or more aspects of the disclosure can be implemented on a computer having a display device, e.g., a CRT (cathode ray tube), LCD (liquid crystal display) monitor, or touch screen for displaying information to the user and optionally a keyboard and a pointing device, e.g., a mouse or a trackball, by which the user can provide input to the computer. Other kinds of devices can be used to provide interaction with a user as well; for example, feedback provided to the user can be any form of sensory feedback, e.g., visual feedback, auditory feedback, or tactile feedback; and input from the user can be received in any form, including acoustic, speech, or tactile input. In addition, a computer can interact with a user by sending documents to and receiving documents from a device that is used by the user; for example, by sending web pages to a web browser on a user's client device in response to requests received from the web browser.

[0061] A number of implementations have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the disclosure. Accordingly, other implementations are within the scope of the following claims.

Claims

1. A computer-implemented method executed on data processing hardware that causes the data processing hardware to perform operations comprising: receiving a training utterance comprising a sequence of audio frames characterizing an utterance and a corresponding ground-truth transcription of the utterance; truncating a suffix portion from the sequence of audio frames to provide a prefix sequence of the audio frames that characterizes a prefix portion of the utterance; processing, using a speech recognition model comprising an encoder and a decoder, the prefix sequence of the audio frames to generate a partial speech recognition hypothesis for the prefix portion of the utterance; processing, using a prefix embedding multi-head attention layer, a sequence of prefix encodings encoded by the encoder of the speech recognition model from the prefix sequence of audio frames to generate a prefix prompt embedding; processing, using a language model (LM), the partial speech recognition hypothesis generated by the speech recognition model conditioned on the prefix prompt embedding to generate a speculated speech recognition hypothesis for the truncated suffix portion of the sequence of audio frames; determining a training loss based on the speculated speech recognition hypothesis and the ground-truth transcription of the utterance; and fine-tuning, using the training loss, parameters of the prefix embedding multi-head attention layer.
2. The computer-implemented method of claim 1, wherein the operations further comprise: obtaining a sequence of trainable query vectors, wherein processing the sequence of prefix encodings encoded by the encoder to generate the prefix prompt embedding comprises processing, using the prefix embedding multi-head attention layer, the sequence of prefix encodings and the sequence of trainable query vectors to generate the prefix prompt embedding.
3. The computer-implemented method of claim 2, wherein: the trainable query vectors comprise soft prompt query vectors; and fine-tuning further comprises fine-tuning the trainable query vectors.
4. The computer-implemented method of claim 1, wherein fine-tuning further comprises

performing parameter-efficient fine-tuning (PEFT) to only update a subset of existing or newly added parameters of the LM based on the training loss.

5. The computer-implemented method of claim 4, wherein: the LM is pre-trained and comprises a plurality of multi-head attention blocks; the pre-trained LM is modified to add two low-rank projection matrices to each multi-head attention block; and performing PEFT comprises fine-tuning only the parameters of the low-rank projection matrices added to the pre-trained LM while existing parameters of the pre-trained LM remain fixed.

6. The computer-implemented method of claim 1, wherein: the encoder of the speech recognition model comprises a pre-trained audio encoder comprising a plurality of multi-head attention layers each comprising a multi-head attention mechanism; and parameters of the pre-trained audio encoder are held fixed.

7. The computer-implemented method of claim 6, wherein the multi-head attention layers comprise Conformer layers.

8. The computer-implemented method of claim 1, wherein the LM comprises a plurality of multi-head attention layers.

9. The computer-implemented method of claim 8, wherein the multi-head attention layers comprise Transformer layers.

10. The computer-implemented method of claim 1, wherein the speech recognition model comprises a recurrent neural network-transducer architecture.

11. A system comprising: data processing hardware; and memory hardware in communication with the data processing hardware and storing instructions that when executed on the data processing hardware causes the data processing hardware to perform operations comprising: receiving a training utterance comprising a sequence of audio frames characterizing an utterance and a corresponding ground-truth transcription of the utterance; truncating a suffix portion from the sequence of audio frames to provide a prefix sequence of the audio frames that characterizes a prefix portion of the utterance; processing, using a speech recognition model comprising an encoder and a decoder, the prefix sequence of the audio frames to generate a partial speech recognition hypothesis for the prefix portion of the utterance; processing, using a prefix embedding multi-head attention layer, a sequence of prefix encodings encoded by the encoder of the speech recognition model from the prefix sequence of audio frames to generate a prefix prompt embedding; processing, using a language model (LM), the partial speech recognition hypothesis generated by the speech recognition model conditioned on the prefix prompt embedding to generate a speculated speech recognition hypothesis for the truncated suffix portion of the sequence of audio frames; determining a training loss based on the speculated speech recognition hypothesis and the ground-truth transcription of the utterance; and fine-tuning, using the training loss, parameters of the prefix embedding multi-head attention layer.

12. The system of claim 11, wherein the operations further comprise: obtaining a sequence of trainable query vectors, wherein processing the sequence of prefix encodings encoded by the encoder to generate the prefix prompt embedding comprises processing, using the prefix embedding multi-head attention layer, the sequence of prefix encodings and the sequence of trainable query vectors to generate the prefix prompt embedding.

13. The system of claim 12, wherein: the trainable query vectors comprise soft prompt query vectors; and fine-tuning further comprises fine-tuning the trainable query vectors.

14. The system of claim 11, wherein fine-tuning further comprises performing parameter-efficient fine-tuning (PEFT) to only update a subset of existing or newly added parameters of the LM based on the training loss.

15. The system of claim 14, wherein: the LM is pre-trained and comprises a plurality of multi-head attention blocks; the pre-trained LM is modified to add two low-rank projection matrices to each multi-head attention block; and performing PEFT comprises fine-tuning only the parameters of the low-rank projection matrices added to the pre-trained LM while existing parameters of the pre-

trained LM remain fixed.

- 16.** The system of claim 11, wherein: the encoder of the speech recognition model comprises a pre-trained audio encoder comprising a plurality of multi-head attention layers each comprising a multi-head attention mechanism; and parameters of the pre-trained audio encoder are held fixed.
 - 17.** The system of claim 16, wherein the multi-head attention layers comprise Conformer layers.
 - 18.** The system of claim 11, wherein the LM comprises a plurality of multi-head attention layers.
 - 19.** The system of claim 18, wherein the multi-head attention layers comprise Transformer layers.
 - 20.** The system of claim 11, wherein the speech recognition model comprises a recurrent neural network-transducer architecture.
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